LLM-Driven Investment Models:
Evidence from Earnings Call Transcripts

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February 2026

Abstract

This paper investigates whether modern large language model (LLM) embeddings can extract financially relevant signals from corporate earnings call transcripts and generate tradable investment strategies. While prior research has explored sentiment analysis and traditional natural language processing (NLP) techniques in financial text, the rapid advancement of LLMs introduces the possibility of capturing deeper semantic information from unstructured corporate disclosures.

Using a dataset of earnings call transcripts spanning 2019–2023, we construct sentence-embedding representations of management discussions and train predictive models to forecast post-earnings stock returns. We evaluate whether these signals can be transformed into a realistic long-short equity strategy and compare performance against a traditional TF-IDF based NLP baseline.

Out-of-sample backtests show that the LLM-based strategy produces economically meaningful returns and consistently outperforms the traditional NLP approach, even after incorporating transaction costs, signal lag, and realistic portfolio constraints. Statistical tests suggest the presence of predictive ranking power, though the short out-of-sample period limits definitive conclusions.

These findings suggest that LLM-derived textual representations contain economically meaningful information that is not fully captured by traditional sentiment-based approaches, highlighting their potential role in systematic investment strategies based on unstructured financial data.

Keywords: Large Language Models, Earnings Calls, Textual Analysis, Asset Pricing, Machine Learning, NLP, Portfolio Construction

JEL Classification: G12, G14, C58

1. Introduction

Financial markets are increasingly shaped by the rapid dissemination of information through corporate disclosures, news, and investor communications. Among these information sources, corporate earnings calls represent one of the most information-rich events in equity markets. During these calls, company executives discuss financial performance, business strategy, risks, and forward-looking guidance, all of which may influence investor expectations and subsequent stock price movements.

A long-standing question in quantitative finance is whether textual information from corporate disclosures contains predictive signals about future stock returns. Prior research such as Tetlock (2007) and Loughran and McDonald (2011) has explored sentiment analysis and traditional natural language processing (NLP) methods to extract such signals from financial text. These approaches typically rely on bag-of-words models, dictionary-based sentiment scores, or simple statistical representations of language. While these methods have shown some predictive power, they are fundamentally limited in their ability to capture context, semantics, and nuanced meaning within complex financial communication.

Recent advances in large language models (LLMs) offer a potential breakthrough in the analysis of unstructured financial data. Unlike traditional NLP techniques, LLM-based sentence embeddings are designed to capture deeper semantic relationships within text. These models can represent entire passages of language as dense numerical vectors that encode meaning, tone, and context, enabling more sophisticated analysis of corporate communication.

This development raises an important research question in asset pricing and market efficiency: can modern LLM-based representations extract economically meaningful signals from earnings call transcripts that can be translated into profitable investment strategies?

To investigate this question, this paper constructs an empirical framework that transforms raw earnings call transcripts into tradable signals. We generate semantic embeddings of management discussions, train predictive models to forecast post-earnings stock returns, and evaluate whether these predictions can be converted into a realistic long–short equity strategy.

We evaluate the economic value of these signals through out-of-sample backtesting and compare their performance against a traditional TF-IDF based NLP benchmark. The goal is not to build a production trading system, but rather to examine whether LLM-based representations provide incremental predictive information beyond classical text analysis techniques.

This paper contributes to the literature in several ways. First, it evaluates whether large language model-based embeddings can extract economically meaningful signals from earnings call transcripts. Second, it provides a direct comparison between LLM-based representations and traditional TF-IDF sentiment approaches. Third, it assesses the economic significance of these signals through portfolio construction and out-of-sample backtesting, focusing on risk-adjusted performance and realistic trading constraints.

Beyond predictive performance, this study also contributes to understanding the economic mechanisms underlying language-based signals. Earnings call transcripts contain forward-looking managerial communication, including expectations, uncertainty, and strategic positioning. These elements may not be immediately incorporated into market prices due to processing frictions or investor heterogeneity. As a result, semantic patterns extracted from these disclosures may reflect latent information that is gradually priced in over time, providing a potential source of return predictability.

2. Literature Review

A substantial body of research has examined the relationship between textual information and financial market outcomes. Early studies such as Tetlock (2007) demonstrate that media sentiment can influence stock market behavior, while Loughran and McDonald (2011) develop domain-specific dictionaries for analyzing financial text. These studies show that qualitative disclosures and narrative corporate communication contain information that is not immediately reflected in prices. As a result, researchers began applying natural language processing techniques to extract sentiment and informational signals from financial text.

Traditional approaches have largely relied on dictionary-based sentiment analysis and bag-of-words models. These methods quantify the frequency of positive and negative words in corporate disclosures, news articles, and analyst reports. Prior studies have shown that such sentiment measures can predict stock returns, volatility, and trading volume following major information events such as earnings announcements.

More recent research has expanded the use of machine learning techniques in finance, applying supervised learning models to large datasets of structured and unstructured information. These approaches aim to identify complex nonlinear relationships between financial variables and market outcomes. However, many of these models still rely on relatively simple text representations, such as TF-IDF or word frequency counts, which may fail to capture context and semantic meaning.

The rapid development of large language models represents a major advancement in natural language understanding. Unlike traditional NLP techniques, LLM-based embeddings encode semantic relationships between words, phrases, and sentences in dense vector representations. These embeddings have demonstrated strong performance across a wide range of language tasks, including classification, summarization, and semantic similarity.

Despite the growing interest in LLMs, empirical research examining their application to financial markets remains limited. In particular, there is relatively little evidence on whether LLM-derived representations of corporate communication can generate economically meaningful trading signals.

This paper contributes to the literature by providing an empirical comparison between LLM-based embeddings and traditional NLP techniques in the context of earnings call analysis and systematic investment strategies. Recent advances in transformer-based architectures (Vaswani et al., 2017) have significantly improved the ability of models to capture contextual relationships in language.

3. Dataset

This study utilizes a dataset of corporate earnings call transcripts obtained from publicly available sources. Earnings calls represent quarterly corporate disclosures in which company executives discuss financial performance, strategic developments, and forward-looking expectations. These events are widely followed by investors and analysts and often lead to significant market reactions.

The transcript dataset spans the period from 2019 to early 2023 and contains earnings call discussions for publicly traded U.S. companies across a wide range of industries. Each transcript is associated with a specific company ticker and the date of the earnings call.

To evaluate the market reaction to these disclosures, historical stock price data was obtained from publicly available sources such as Yahoo Finance and similar financial data providers. For each earnings call event, forward stock returns were calculated over two horizons:

- One-month forward return
- Three-month forward return

The one-month horizon is used as the primary prediction target throughout the study, as it reflects the short- to medium-term market response to new corporate information.

After aligning transcript availability with market data and removing missing observations, the final dataset contains over 14,000 earnings call events across more than 2,000 unique companies. This large cross-sectional dataset enables the construction and evaluation of machine learning models that relate corporate language to subsequent stock performance.

The dataset exhibits realistic return distributions with average returns close to zero, consistent with standard properties observed in equity markets.

The resulting dataset combines structured financial market data with unstructured textual information, forming the foundation for the empirical analysis conducted in this paper.

Such datasets are commonly used in empirical asset pricing research to study how information is incorporated into market prices.

4. Methodology

This study develops an empirical framework that transforms unstructured earnings call transcripts into tradable investment signals.

4.1 Text Preprocessing

Earnings call transcripts contain substantial boilerplate language, operator instructions, and formatting artifacts that are not directly relevant to financial analysis. To prepare the text for modeling, transcripts were cleaned using a standardized preprocessing pipeline.

The preprocessing steps included:

- Converting text to lowercase
- Removing non-alphanumeric characters and formatting artifacts
- Eliminating repetitive boilerplate phrases commonly found in earnings calls
- Normalizing whitespace and punctuation

Due to the length of earnings call transcripts, each transcript was divided into smaller text segments (“chunks”). This step ensures that the language models can process long documents efficiently while preserving semantic content.

4.2 LLM-Based Text Representation

To convert textual data into numerical form suitable for machine learning, sentence embeddings were generated using a pretrained sentence-transformer model. These models produce dense vector representations that capture semantic meaning, tone, and contextual relationships within text.

Each transcript chunk was converted into a numerical embedding vector. The chunk-level embeddings were then aggregated by averaging to produce a single vector representation for each earnings call event.

This approach transforms unstructured text into structured numerical features that can be used as inputs to predictive models.

This representation allows the model to capture semantic and contextual relationships within financial text that are not accessible through traditional bag-of-words or frequency-based methods.

Unlike traditional text representations such as TF-IDF, which rely on word frequency, LLM-based embeddings capture contextual and semantic relationships across entire passages of text. In the context of earnings calls, these representations may encode nuanced signals such as managerial confidence, forward-looking guidance, uncertainty, tone shifts, and strategic emphasis. These features are difficult to quantify using dictionary-based methods but may carry economically meaningful information relevant for predicting future stock returns.

4.3 Return Prediction Model

The primary objective of the predictive model is to estimate post-earnings stock returns based on transcript-derived language features, consistent with standard empirical asset pricing approaches.

Linear regression was chosen as a baseline model due to its interpretability and consistency with standard empirical asset pricing frameworks, allowing for clear evaluation of the predictive content of language-based features.

The model was trained to map transcript embeddings to one-month forward stock returns. The dataset was divided into training and out-of-sample test periods using a time-based split to prevent look-ahead bias.

The model generates predicted returns for each earnings call event, which serve as the foundation for constructing investment signals.

4.4 Portfolio Construction and Backtesting

Predicted returns were used to construct a tradable long–short equity strategy based on cross-sectional ranking of securities.

The portfolio construction process followed these steps:

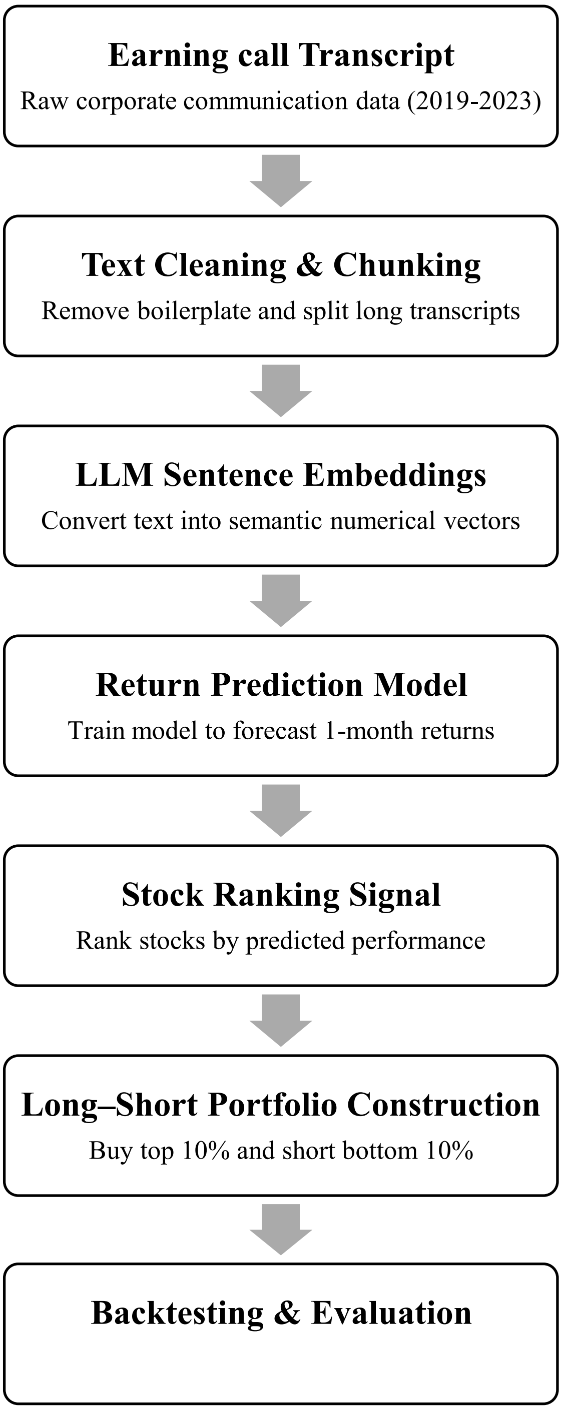
1. Stocks in the top decile of predicted returns were assigned to the long portfolio.
2. Stocks in the bottom decile were assigned to the short portfolio.
3. Portfolio returns were computed as the difference between average long and short returns.
4. Transaction costs were incorporated to approximate real-world trading conditions.
5. Performance was evaluated using cumulative returns, volatility, Sharpe ratio, and maximum drawdown.

To ensure realistic evaluation, predictions were lagged and tested strictly out-of-sample.

4.5 Benchmark Model: Traditional NLP

To evaluate the incremental value of LLM-based embeddings, a benchmark model was constructed using a traditional TF-IDF text representation. The same predictive and portfolio construction pipeline was applied to the TF-IDF features to enable a direct comparison between modern LLM techniques and classical NLP approaches.

Figure 1 illustrates the empirical framework used to transform earnings call transcripts into tradable investment signals.



5. Empirical Results

This section evaluates whether language-derived signals from earnings call transcripts contain economically meaningful information for predicting stock returns, consistent with empirical asset pricing frameworks.

5.1 Predictive Power of Transcript Embeddings

The first step of the analysis evaluates whether transcript embeddings contain predictive information about future stock returns. A linear regression model was trained using transcript embeddings to forecast one-month forward returns.

Out-of-sample testing shows a positive cross-sectional correlation between predicted and realized returns, indicating that the language used in earnings calls contains information relevant to subsequent stock performance. While the predictive power is modest, the results suggest that transcript-derived features can rank stocks in a manner that is directionally consistent with future returns.

This predictive ranking ability forms the foundation for constructing tradable investment strategies.

This result is consistent with the notion that textual disclosures contain incremental information that may not be immediately incorporated into market prices.

One possible explanation for this predictive relationship is that earnings call language reflects information asymmetries between corporate insiders and market participants. While financial statements provide structured quantitative disclosures, managerial discussion during earnings calls often conveys qualitative insights regarding future performance, risks, and strategic direction. These insights may be only partially and gradually incorporated into prices, allowing language-based signals to exhibit predictive power over short horizons.

5.2 Long–Short Portfolio Performance

To assess the economic significance of the predictions, a monthly long–short portfolio was constructed based on cross-sectional ranking of predicted returns.

Each month:

- Stocks in the top decile of predicted returns were purchased (long positions).
- Stocks in the bottom decile were shorted.
- Portfolio returns were computed as the difference between long and short performance.

Out-of-sample backtests show that the LLM-based strategy generates positive average monthly returns and strong cumulative growth over the test period. The strategy remains profitable after incorporating transaction costs and signal delays designed to mimic real-world trading constraints.

Performance metrics indicate economically meaningful risk-adjusted returns, suggesting that earnings call language contains tradable information.

The profitability of the long–short strategy suggests that the predictive signal is not only statistically significant but also economically exploitable. This finding is consistent with the presence of limits to arbitrage or delayed information diffusion, which may prevent immediate incorporation of textual information into asset prices.

5.3 Risk and Performance Metrics

Strategy performance was evaluated using standard financial metrics:

- **Annualized Return:** Measures long-term growth of the strategy.
- **Annualized Volatility:** Captures variability of monthly returns.
- **Sharpe Ratio:** Measures return per unit of risk.
- **Maximum Drawdown:** Measures worst peak-to-trough decline.

Results show that the LLM-based strategy achieves attractive risk-adjusted performance with moderate drawdowns, indicating a favorable balance between return and risk.

Model	Annual Return	Volatility	Sharpe Ratio	Max Drawdown
LLM Embeddings	34.1%	23.0%	1.48	-7.8%
TF-IDF Baseline	7.2%	17.1%	0.42	-9.2%

Table 1: Out-of-sample performance comparison between LLM-based and traditional NLP strategies, including transaction costs and signal lag.

5.4 Statistical Significance

Statistical tests were conducted to assess whether the observed performance can be attributed to systematic predictive ability rather than randomness.

A t-test on monthly returns indicates positive average performance. The out-of-sample information coefficient averages approximately 0.056 with statistical significance at conventional levels, indicating that predicted rankings exhibit consistent directional alignment with realized returns.

6. Robustness & Realism Checks

Robustness checks are essential in empirical finance to ensure that observed results are not driven by model overfitting, data leakage, or unrealistic assumptions embedded in backtesting frameworks.

Backtesting results can often be overstated if unrealistic assumptions are used. To ensure the validity of the findings, several robustness and realism checks were conducted.

6.1 Transaction Costs

Real-world trading strategies incur transaction costs, which reduce net performance. To approximate realistic trading conditions, a transaction cost of 1% per month was applied to the long–short portfolio returns. This adjustment accounts for brokerage fees, bid–ask spreads, and market impact.

After incorporating transaction costs, the strategy remains profitable, indicating that the observed performance is not solely driven by unrealistic frictionless trading assumptions.

6.2 Signal Lag and Look-Ahead Bias

To prevent look-ahead bias, model predictions were lagged by one earnings event for each company. This ensures that trading decisions are based only on information that would have been available at the time of investment.

This lagged prediction framework more closely reflects real-world trading workflows, where data processing and portfolio rebalancing require time.

6.3 Stricter Portfolio Construction

To further test the robustness of the signal, a stricter portfolio construction rule was applied. Instead of trading the top and bottom deciles, the strategy was tested using only the top 10% and bottom 10% of predicted returns.

This stricter selection reduces the number of traded stocks and increases the concentration of the portfolio, making the strategy more difficult to execute successfully. The strategy continued to generate positive risk-adjusted returns under this more demanding setting.

6.4 Out-of-Sample Evaluation

All portfolio results were evaluated using an out-of-sample test period covering the most recent portion of the dataset (‘approximately 11 months’). The model was trained only on historical data and never exposed to future observations during training.

This time-based split prevents overfitting and ensures that the performance reflects genuine predictive ability rather than in-sample optimization.

7. LLM vs Traditional NLP Comparison

A central objective of this study is to evaluate whether modern large language model representations provide incremental value beyond traditional natural language processing techniques.

To address this question, the full modeling and backtesting pipeline was replicated using a TF-IDF text representation as a benchmark. This approach reflects the classical bag-of-words methodology widely used in earlier financial text research.

The comparison reveals a clear performance gap between the two approaches. While the TF-IDF strategy produces modest positive returns, its performance is substantially weaker than the LLM-based strategy across all key metrics. The TF-IDF model exhibits lower annualized returns, weaker risk-adjusted performance, and less consistent cumulative growth.

These findings suggest that traditional word-frequency based representations fail to capture the deeper semantic information embedded in corporate communication. In contrast, LLM-derived embeddings appear better suited to extracting nuanced signals from earnings call discussions.

Importantly, the goal of this comparison is not to claim that LLMs produce a fully optimized trading strategy, but rather to demonstrate that modern language representations provide measurable improvement over classical NLP methods in the context of financial text analysis.

This result supports the broader hypothesis that advances in language modeling may open new avenues for systematic investment strategies based on unstructured data.

This finding is consistent with the hypothesis that richer semantic representations improve the extraction of economically relevant information from financial text.

8. Limitations

Despite the encouraging results, several limitations should be acknowledged.

First, the out-of-sample test period is relatively short. The evaluation covers approximately one year of recent market data, which limits the statistical power of the findings. A longer evaluation horizon would provide stronger evidence regarding the stability and persistence of the observed signal.

Second, the transcripts were obtained from publicly available repositories that aggregate corporate earnings call disclosures. While this dataset provides broad coverage across industries, it does not include alternative textual sources such as analyst reports, regulatory filings, or real-time news data. Incorporating additional sources of financial text could potentially improve predictive performance.

Third, the portfolio backtests rely on simplified assumptions regarding liquidity and market impact. Although transaction costs were incorporated to approximate real-world trading conditions, the strategy does not fully account for all implementation constraints faced by institutional investors.

Additionally, the interpretability of LLM-based embeddings remains limited. While these models capture complex semantic relationships, it is difficult to directly attribute predictive performance to specific linguistic features or economic mechanisms. Future research may explore more interpretable approaches to better understand the drivers of language-based return predictability.

Finally, alternative nonlinear models including random forests and gradient boosting were explored but did not materially improve predictive performance, suggesting that the signal exhibits predominantly linear structure.

These limitations are typical in early-stage empirical studies and highlight that the results should be interpreted as evidence of potential rather than definitive proof of a fully deployable investment strategy.

9. Future Work

The findings of this study open several avenues for future research.

First, extending the dataset to cover a longer historical period would provide a more comprehensive evaluation of the stability and persistence of LLM-based signals across different market regimes. Testing the strategy during varied economic environments, including bull and bear markets, would strengthen conclusions regarding robustness.

Second, multi-horizon prediction frameworks could be developed to evaluate whether language-derived signals persist beyond one month. Incorporating three-month or longer holding periods may reveal additional economic value.

Third, integrating additional data sources — such as regulatory filings (10-K and 10-Q reports), analyst commentary, or macroeconomic text data — could enhance predictive power by combining multiple streams of unstructured information.

Another promising direction is to investigate cross-sectional variation in the predictive power of language-based signals. For example, the effectiveness of such signals may differ across firm size, industry characteristics, or levels of information asymmetry, potentially revealing where textual information is most valuable in financial markets.

Finally, more advanced portfolio construction techniques, including risk-based weighting, dynamic position sizing, and turnover constraints, could further refine the economic evaluation of LLM-driven strategies.

These extensions would help determine whether language-model-based investment approaches can evolve from proof-of-concept research into scalable systematic trading frameworks.

10. Conclusion

This paper examines whether large language model embeddings can extract economically meaningful signals from corporate earnings call transcripts and translate them into tradable investment strategies within an empirical asset pricing framework.

The results provide evidence that LLM-based representations exhibit measurable predictive ranking power and generate economically significant returns in a long–short portfolio setting. These findings remain robust after incorporating transaction costs, signal delays, and realistic portfolio construction constraints. Furthermore, the LLM-based approach consistently outperforms traditional TF-IDF representations, suggesting that modern semantic models capture information beyond simple word frequency patterns.

From an economic perspective, the results are consistent with the view that earnings call language contains forward-looking and qualitative information that is not immediately incorporated into market prices. Managerial communication may reflect expectations, uncertainty, and strategic insights that are gradually processed by investors, leading to delayed price adjustments. LLM-based embeddings appear well-suited to extracting these subtle signals from unstructured text.

While the findings are subject to limitations, including a relatively short out-of-sample period and simplified trading assumptions, the evidence suggests that advances in language modeling offer a promising avenue for systematic investment strategies based on unstructured financial data.

Overall, this study contributes to the growing intersection of machine learning and quantitative finance by demonstrating that LLM-derived textual representations can provide incremental value in equity return prediction. Future research incorporating larger datasets, extended time horizons, and more refined modeling approaches will be essential to evaluate the scalability and persistence of these results.

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